Partial derivatives and slices

Partial derivative: vary one input while holding the others fixed. For a scalar-valued f of n variables and the ith standard basis vector $\mathbf{e}_i$,

$$f_{x_i}(\mathbf{a}) = \frac{\partial f}{\partial x_i}(\mathbf{a}) = \lim_{h\to 0} \frac{f(\mathbf{a}+h\mathbf{e}_i) - f(\mathbf{a})}{h},$$

provided the limit exists. Other notations are $\partial_{x_i} f$, $D_{x_i} f$, and $D_i f$.

Two variables:

$$f_x(a,b) = \lim_{h\to 0} \frac{f(a+h,b) - f(a,b)}{h}, \qquad f_y(a,b) = \lim_{h\to 0} \frac{f(a,b+h) - f(a,b)}{h}.$$

Slice: fixing $y = b$ gives $g(x) = f(x,b)$; fixing $x = a$ gives $g(y) = f(a,y)$, on the corresponding domains.

Geometric interpretation: $f_x(a,b)$ and $f_y(a,b)$ are the tangent slopes of the graph's slices in the planes $y = b$ and $x = a$, respectively, when these derivatives exist.

Higher-order partial derivatives

Order: subscripts run left to right; differential operators act right to left:

$$f_{x_i x_j} = (f_{x_i})_{x_j} = \frac{\partial^2 f}{\partial x_j\, \partial x_i} = D_j D_i f.$$

Second partials in two variables: $f_{xx} = \partial^2 f/\partial x^2$, $f_{yy} = \partial^2 f/\partial y^2$, $f_{xy} = (f_x)_y = \partial^2 f/(\partial y\, \partial x)$, and $f_{yx} = (f_y)_x = \partial^2 f/(\partial x\, \partial y)$.

Clairaut's theorem: if f is defined on an open ball B containing $\mathbf{a}$, and $f_{x_i x_j}$ and $f_{x_j x_i}$ exist and are continuous on B, then $f_{x_i x_j}(\mathbf{a}) = f_{x_j x_i}(\mathbf{a})$. Without these hypotheses, mixed partials need not agree.

Gradient and Hessian

Gradient: where all first partials exist, the column vector

$$\nabla f(\mathbf{x}) = \begin{bmatrix} f_{x_1}(\mathbf{x}) \\ \vdots \\ f_{x_n}(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^n.$$

Hessian: where all second partials exist, the $n \times n$ matrix with entry $(H_f)_{ij} = f_{x_i x_j}$. Row i differentiates f_{x_i} in each input variable. In two variables,

$$H_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

Symmetry: continuous mixed second partials on a ball around $\mathbf{a}$ imply $H_f(\mathbf{a})^T = H_f(\mathbf{a})$ by Clairaut's theorem.

Jacobian matrices

Jacobian: for $F : \mathbb{R}^n \to \mathbb{R}^m$, where all first component partials exist, J_F is the $m \times n$ matrix with $(J_F)_{ij} = \partial F_i/\partial x_j$. Rows index output components; columns index input coordinates.

Identities and shapes: $J_f = (\nabla f)^T$ $(1 \times n)$, ∇f $(n \times 1)$, and $J_{\nabla f} = H_f$ $(n \times n)$.